DS HOMEWORK PROGRAMS EXERCISE-3

1.Write a C program to store n elements in an array using dynamic memory allocation and find largest number.

```
#include <stdio.h>
#include <stdlib.h>
int main() {
   int n, i, largest;
   int *arr;
   printf("Enter the number of elements: ");
   scanf("%d", &n);
   arr = (int *)malloc(n * sizeof(int));
   printf("Enter %d elements:\n", n);
   for (i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }
   largest = arr[0];
   for (i = 1; i < n; i++) {
      if (arr[i] > largest) {
         largest = arr[i];
      }
   }
   printf("Largest number = %d\n", largest);
   free(arr);
   return 0;
}
```

OUTPUT

```
Enter the number of elements: 5
Enter 5 elements:
12
45
7
```

89

23

Largest number = 89

2. Print an array in reverse order using pointers.

```
#include <stdio.h>
int main() {
   int n, i;
   int arr[100];
   int *ptr;
   printf("Enter the number of elements: ");
   scanf("%d", &n);
   printf("Enter %d elements:\n", n);
   for (i = 0; i < n; i++) {
      scanf("%d", &arr[i]);
   }
   ptr = &arr[n - 1];
   printf("Array in reverse order:\n");
   for (i = n - 1; i >= 0; i--) {
      printf("%d ", *ptr);
      ptr--;
   }
   return 0;
}
```

OUTPUT

Enter the number of elements: 5

Enter 5 elements:

10

20

30

40

50

Array in reverse order:

50 40 30 20 10

3. Write a C program to store information of students using structure, sort them by Name.

```
#include <stdio.h>
#include <string.h>
struct Student {
    char name[50];
    int roll;
    float marks;
};
int main() {
    int n, i, j;
    struct Student s[100], temp;
    printf("Enter the number of students: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("Enter name, roll number and marks for student %d:\n", i + 1);
        scanf("%s %d %f", s[i].name, &s[i].roll, &s[i].marks);
    }
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (strcmp(s[i].name, s[j].name) > 0) {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }
    printf("\nStudents sorted by Name:\n");
```

```
    for (i = 0; i < n; i++) {
        printf("Name: %s, Roll No: %d, Marks: %.2f\n",
               s[i].name, s[i].roll, s[i].marks);
    }
    return 0;
}
```

OUTPUT

```
Enter the number of students: 3
Enter name, roll number and marks for student 1:
Alice 101 85.5
Enter name, roll number and marks for student 2:
Jack 102 91.0
Enter name, roll number and marks for student 3:
Bob 103 78.5
Students sorted by Name:
Name: Alice, Roll No: 101, Marks: 85.50
Name: Bob, Roll No: 103, Marks: 78.50
Name: Jack, Roll No: 102, Marks: 91.00
```

4. Write a C program to add 'n' distances given in inch-feet system using structures.

```
#include <stdio.h>
struct Distance {
    int feet;
    float inch;
};
int main() {
    int n, i;
    struct Distance d[100], sum = {0, 0};
    printf("Enter the number of distances: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
```

```
        printf("Enter distance %d (feet inches): ", i + 1);
        scanf("%d %f", &d[i].feet, &d[i].inch);
        sum.feet += d[i].feet;
        sum.inch += d[i].inch;
    }
    if (sum.inch >= 12) {
        sum.feet += (int)(sum.inch / 12);
        sum.inch = (int)sum.inch % 12;
    }
    printf("Total distance = %d feet %.2f inches\n", sum.feet, sum.inch);
    return 0;
}
```

Output

```
Enter the number of distances: 3
Enter distance 1 (feet inches): 5 8
Enter distance 2 (feet inches): 3 7
Enter distance 3 (feet inches): 2 10
Total distance = 12 feet 1.00 inches
```

5. Write a C program to add two complex numbers by passing structure to a function.

```
#include <stdio.h>
struct Complex {
    float real;
    float imag;
};
struct Complex add(struct Complex c1, struct Complex c2) {
    struct Complex result;
    result.real = c1.real + c2.real;
    result.imag = c1.imag + c2.imag;
    return result;
}
```

```
int main() {
    struct Complex c1, c2, sum;
    printf("Enter real and imaginary parts of first complex number: ");
    scanf("%f %f", &c1.real, &c1.imag);
    printf("Enter real and imaginary parts of second complex number: ");
    scanf("%f %f", &c2.real, &c2.imag);
    sum = add(c1, c2);
    printf("Sum = %.2f + %.2fi\n", sum.real, sum.imag);
    return 0;
}
```

Output

```
Enter real and imaginary parts of first complex number: 3 4
Enter real and imaginary parts of second complex number: 5 2
Sum = 8.00 + 6.00i
```

6. Write a C program to maintain inventory management system using structure

```
#include <stdio.h>
struct Product {
    int id;
    char name[50];
    int quantity;
    float price;
};
int main() {
    int n, i;
    struct Product p[100];
    printf("Enter the number of products: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("\nEnter details of product %d:\n", i + 1);
        printf("Product ID: ");
```

```
        scanf("%d", &p[i].id);
        printf("Product Name: ");
        scanf("%s", p[i].name);
        printf("Quantity: ");
        scanf("%d", &p[i].quantity);
        printf("Price: ");
        scanf("%f", &p[i].price);
    }
    printf("\nInventory Details:\n");
    for (i = 0; i < n; i++) {
        printf("\nProduct ID: %d\n", p[i].id);
        printf("Product Name: %s\n", p[i].name);
        printf("Quantity: %d\n", p[i].quantity);
        printf("Price: %.2f\n", p[i].price);
        printf("Total Value: %.2f\n", p[i].quantity * p[i].price);
    }
    return 0;
}
```

Output

```
Enter the number of products: 3

Enter details of product 1:
Product ID: 101
Product Name: Pen
Quantity: 50
Price: 10

Enter details of product 2:
Product ID: 102
Product Name: Book
Quantity: 20
```

Price: 50

Enter details of product 3:

Product ID: 103

Product Name: Pencil

Quantity: 100

Price: 5

Inventory Details:

Product ID: 101

Product Name: Pen

Quantity: 50

Price: 10.00

Total Value: 500.00

Product ID: 102

Product Name: Book

Quantity: 20

Price: 50.00

Total Value: 1000.00

Product ID: 103

Product Name: Pencil

Quantity: 100

Price: 5.00

Total Value: 500.00

7. Write a C program for decimal to binary, octal and hexa decimal conversion.

#include <stdio.h>

```
int main() {
   int n;
   printf("Enter a decimal number: ");
   scanf("%d", &n);
   printf("Binary = ");
   int temp = n, binary[32], i = 0;
   while (temp > 0) {
      binary[i++] = temp % 2;
      temp /= 2;
   }
   while (i > 0)
      printf("%d", binary[--i]);
   printf("\nOctal = %o", n);
   printf("\nHexadecimal = %X\n", n);
   return 0;
}
```

Output

```
Enter a decimal number: 25
Binary = 11001
Octal = 31
Hexadecimal = 19
```

8.Write a C program to calculate difference between two time periods.

```
#include <stdio.h>
struct Time {
   int hours;
   int minutes;
   int seconds;
};
int main() {
   struct Time t1, t2, diff;
```

```
    printf("Enter first time (hours minutes seconds): ");
    scanf("%d %d %d", &t1.hours, &t1.minutes, &t1.seconds);
    printf("Enter second time (hours minutes seconds): ");
    scanf("%d %d %d", &t2.hours, &t2.minutes, &t2.seconds);
    if (t2.seconds < t1.seconds) {
        t2.seconds += 60;
        t2.minutes--;
    }
    if (t2.minutes < t1.minutes) {
        t2.minutes += 60;
        t2.hours--;
    }
    diff.hours = t2.hours - t1.hours;
    diff.minutes = t2.minutes - t1.minutes;
    diff.seconds = t2.seconds - t1.seconds;
    printf("Difference = %d:%d:%d\n",
           diff.hours, diff.minutes, diff.seconds);
    return 0;
}
```

Output

```
Enter first time (hours minutes seconds): 5 45 30
Enter second time (hours minutes seconds): 8 20 15
Difference = 2:34:45
```

9. Write a C program to implement an employee management system using structures.

```
#include <stdio.h>
struct Employee {
    int id;
    char name[50];
    float salary;
};
```

```
int main() {
    int n, i;
    struct Employee emp[100];
    printf("Enter the number of employees: ");
    scanf("%d", &n);
    for (i = 0; i < n; i++) {
        printf("\nEnter details of employee %d:\n", i + 1);
        printf("Employee ID: ");
        scanf("%d", &emp[i].id);
        printf("Employee Name: ");
        scanf("%s", emp[i].name);
        printf("Salary: ");
        scanf("%f", &emp[i].salary);
    }
    printf("\nEmployee Details:\n");
    for (i = 0; i < n; i++) {
        printf("\nEmployee ID: %d\n", emp[i].id);
        printf("Employee Name: %s\n", emp[i].name);
        printf("Salary: %.2f\n", emp[i].salary);
    }
    return 0;
}
```

Output

```
Enter the number of employees: 3

Enter details of employee 1:
Employee ID: 101
Employee Name: Alice
Salary: 45000
```

Enter details of employee 2:

Employee ID: 102

Employee Name: Bob

Salary: 50000

Enter details of employee 3:

Employee ID: 103

Employee Name: Jack

Salary: 40000

Employee Details:

Employee ID: 101

Employee Name: Alice

Salary: 45000.00

Employee ID: 102

Employee Name: Bob

Salary: 50000.00

Employee ID: 103

Employee Name: Jack

Salary: 40000.00

10. Write a C program to calculate area and perimeter of a rectangle and triangle using function pointers.

```
#include <stdio.h>
#include <math.h>
float rectangleArea(float l, float b) {
   return l * b;
}
```

```
float rectanglePerimeter(float l, float b) {
   return 2 * (l + b);
}
float triangleArea(float b, float h) {
   return 0.5 * b * h;
}
float trianglePerimeter(float a, float b, float c) {
   return a + b + c;
}

int main() {
   float l, b, h, a, c;
   float (*area)(float, float);
   float (*perimeter)(float, float);
   printf("Enter length and breadth of rectangle: ");
   scanf("%f %f", &l, &b);
   area = rectangleArea;
   perimeter = rectanglePerimeter;
   printf("Rectangle Area = %.2f\n", area(l, b));
   printf("Rectangle Perimeter = %.2f\n", perimeter(l, b));
   printf("\nEnter three sides of triangle: ");
   scanf("%f %f %f", &a, &b, &c);
   printf("Enter height of triangle: ");
   scanf("%f", &h);

   area = triangleArea;
   printf("Triangle Area = %.2f\n", area(b, h));
   printf("Triangle Perimeter = %.2f\n", trianglePerimeter(a, b, c));
   return 0;
}
```

Output

Enter length and breadth of rectangle: 10 5

Rectangle Area = 50.00

Rectangle Perimeter = 30.00

Enter three sides of triangle: 3 4 5

Enter height of triangle: 4

Triangle Area = 8.00

Triangle Perimeter = 12.00